

CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy

Gregory Schwing^{a)}

Department of Surgery, Detroit Medical Center and Wayne State University, Detroit, Michigan, USA

Ashley Schehr, Annika Tekumulla, Margret Khoushi, Ryan Christian, Dane Hubers, Faris Mahjoub, Hassan Saad, Mia Sooch, Sathyagopal Siddapureddy, Michael McLellan, and Jerick Kim

School of Medicine, Wayne State University, Detroit, Michigan, USA

Miraziz Ismoilov and Nizar Alnabahneh

Department of Radiology, Detroit Medical Center and Wayne State University, Detroit, Michigan, USA

(Dated: 13 September 2026)

Purpose: The gold standard for naming a vertebra at the lumbosacral junction is counting caudally from C2 on whole-spine imaging. In practice, a lumbar surgical case is planned on lumbar-only imaging (T12 to S1) without C2, making the gold standard method impossible. Abdominopelvic computed tomography (CT) holds the lumbar-only imaging span plus the lowest ribs and pelvis, making it a reasonable substitute for the preoperative view. When a lumbosacral transitional vertebra (LSTV) alters the count, however, the local anatomy is much more ambiguous. A person who has four rib-free lumbar vertebrae may have an L1 with a lumbar rib or an L5 assimilated to the sacrum, and a person who has six rib-free lumbar vertebrae may have an L6, a T12 with aplastic ribs, or a lumbarized S1. CTSpinoPelvic1K permits asking whether local morphology can resolve that ambiguity without counting caudally from C2. Two public label sets already covered this imaging, CTSpine1K’s vertebrae and CTPelvic1K’s pelvis, on the same colonography patients. This release joins them in one dataset of 802 CT records and adds ribs, femora and surgical hardware. It also introduces classes for L6, T13 and lumbar ribs, a class vocabulary that describes more of the anomalies at this junction than any existing public CT collection.

Acquisition and Validation Methods: Records pair CTSpine1K and CTPelvic1K masks on each patient’s bone-richest TCIA acquisition, and the release carries the series keys that map every mask to its volume. Validation covered geometric invariants (802/802 pass), rib-vertebra incidence across 5,749 ribs (0.035% offset), and spinopelvic measures matching published values.

Data Format and Usage Notes: NIfTI image/label pairs with patient-grouped LSTV-stratified five-fold splits and a loader; archived at <https://doi.org/10.5281/zenodo.22139642>.

Potential Applications: Classifying a vertebra from local features; updating cadaveric morphometry; spinopelvic assessment; opportunistic screening; imaging-based prevalence of anatomical variants in an asymptomatic cohort; and spinopelvic biomechanical modelling from patient-matched prone and supine acquisitions. *Limitations:* thoracic ground truth is field-of-view limited; postural angles are non-standing; no held-out test set; ribs are triaged-review pseudolabels; Castellvi grades are two-reader consensus on 33 records.

I. INTRODUCTION

The spine typically has seven cervical, twelve thoracic, and five lumbar vertebrae. A vertebra is classified by where it falls in that sequence. Transitional variants sit at the two ends of the lumbar column, with the thoracolumbar transitional vertebra (TLTV) above and the lumbosacral transitional vertebra (LSTV) below. Both transitional variants disturb identification by changing what the border vertebra and its ribs *look like*. For example, a thoracolumbar border vertebra may carry a full rib, a hypoplastic stump, or none at all, and a

lumbosacral one may be partly or completely assimilated to the sacrum (*sacralization*), with an anteriorly wedged body, a reduced disc beneath it and hypoplastic facet joints, or separated from it (*lumbarization*), with a squared body, lumbar-type facets and a full-height disc¹. In addition, they change how many vertebrae each region holds: eleven or thirteen rib-bearing thoracic and four or six rib-free lumbar vertebrae.

Together, these transitional variants leave the vertebra at each border without a definitive name. The one formal nomenclature at the lumbosacral junction, Castellvi’s,

grades the morphology of the transverse process and says nothing about whether the segment is L5 or S1^{1,2}; every imaging landmark proposed for numbering has proved unreliable in transitional cases³⁻⁵; and even a count from C2 yields a position, which a convention then converts to a name. Each of the readings above yields the same count (Fig. 2), and the morphology that would separate them is not always decisive to human readers. Whether it differs enough for a trained classifier to separate them, or at least to assign each reading a probable classification, is the question this release aims to answer. The gold standard for numeration in transitional anatomy is whole-spine imaging, counting caudally from C2⁶, which determines how many vertebrae and how many rib pairs the column holds. No spine-limited acquisition can supply either count. LSTV is common, reported at 10–29% in the general population⁷ and in 16.3% of a whole-spine CT series⁸, and wrong-level surgery is its most serious consequence.

Existing public collections cannot address this, and not for lack of size (Table 1). Prior spine datasets omit the pelvis and pelvic datasets do not number the vertebrae. TotalSegmentator⁹ has no class for a sixth lumbar vertebra or for a rib on a lumbar vertebra, and VerSe¹⁰, the one collection with an L6 class, stops at the sacrum and carries no ribs. As such, none of these collections can record a spine with six lumbar vertebrae together with both of its borders, and a segmenter trained on them must shift the whole column by a level or absorb a vertebra into its neighbor when it encounters an enumeration anomaly. This problem has now been measured on CT, where prior labeling methods assigned every vertebra correctly in 77% of subjects and an anomaly-aware extension of SPINEPS¹¹ raised that to 99%¹². Its weights label L6 and T13 on CT, but its 1,536-scan CT cohort is in-house and unreleased, so what exists is a capability, not a dataset.

When considering the intraoperative side, LevelCheck registers the intraoperative radiograph to the preoperative CT and projects the CT’s vertebral labels onto it¹³⁻¹⁵. Those labels, however, are placed by hand and verified by the surgeon¹³, so the method assumes a correctly labeled CT and does not supply one itself. As such, a dataset carrying the anomaly classes is vital for automating that step.

CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and classifies each structure an enumeration anomaly produces, including a sixth lumbar vertebra and lumbar ribs. A scheme without those classes is unable to adequately describe anatomic anomalies.

The project began with the question whether a vertebra can be named from the anatomy on the images a lumbar surgical case is actually planned on. By guideline those images are lumbar-only¹⁶⁻¹⁹ and whole-spine coverage is reserved for trauma with an identified injury^{20,21} and for deformity radiographs. An abdominopelvic CT runs from the diaphragm to the pelvic floor, so it cap-

tures that span with the lowest ribs and the whole pelvis, while lacking C2. Every record comes from one prospective trial protocol, ACRIN 6664^{22,23}, which scanned 802 patients aged 50 and over, supine and prone, on multi-detector CT at 15 centers, on five scanner vendors and nine models, with manufacturer, model and reconstruction kernel recorded per record. To our knowledge, it is the largest spine-and-pelvis annotated CT cohort acquired under a single protocol. The comparators, by contrast, pool collections, are multi-site by design, or are routine clinical scans under many protocols (Table 1). With the protocol for this study fixed, scanner effects on a model become measurable rather than confounded. The downside is the restriction to an abdominal field of view, in which only the lowest thoracic levels are present (Sec. V).

The gap was evident. CTPelvic1K²⁴ and CTSpine1K²⁵ each annotated the COLONOG collection under radiologist supervision for the same patients, with one annotating the pelvis and the other annotating the vertebrae. The two had remained separate, although joining them needs no new imaging and no new radiologist. It was not a file merge, because each annotation had to be traced back to the series it was drawn on (Sec. II.A). Once completed, that merge yielded a combined spine-and-pelvis frame for 802 patients, and the bones neither set had, ribs, femora and hardware, could be annotated against it rather than from scratch.

I.A. Vertebra labeling when the scan cannot settle the count

The main limitation is that no scan in this dataset contains C2, so the gold standard conventional count cannot be performed. This is a property of abdominal imaging rather than of the annotation itself. A thirteenth thoracic vertebra and an L1 with a lumbar rib are different phenotypes, distinguished as separate subtypes in cadaveric classifications of the thoracolumbar junction^{26,27}. The first is an additional rib-bearing vertebra positioned above five lumbar vertebrae, so the column has 25 presacral vertebrae. The second is a rudimentary rib on a lumbar-type L1 in a column with the standard 24 presacral vertebrae. Neither is separable by counting from the last rib-bearing vertebra or from the sacrum because four rib-free vertebrae can fit a lumbar rib, a cranially shifted T13 or an L5 assimilated to the sacrum, and six rib-free vertebrae can fit an aplastic twelfth rib or a sixth lumbar vertebra. A *stump rib* is a hypoplastic rib that is a key indicator of a thoracolumbar transitional vertebra. On whole-spine CT scans, rib anomalies travel with the lumbosacral border and hypoplastic twelfth ribs occur with sacralization and lumbar ribs with lumbarization.⁸

In this release, every vertebra carries the identifier its radiologist-sourced annotation assigned as the ground truth, corrected where the source was wrong. In addition, every released measurement is expressed against the two landmarks present in essentially every abdominal CT

scan: the sacrum inferiorly and the lowest rib-bearing vertebra superiorly. This makes the measurements positional rather than nominal, with a vertebra being classified by how many rib-free vertebrae separate it from the sacrum, a lowest rib by its length as a fraction of the rib above it, and a transverse process by its span and its distance from the ala. None of these values shift with the name a reader gives the junction, meaning that cases that disagree about the name remain comparable.

But the vertebrae themselves are known to differ. Level-specific morphometry is long established in the literature: pedicle width and height change systematically from the thoracic to the lumbar spine,²⁸ vertebral body, endplate and canal dimensions differ by level,^{29–31} and the transverse process and the iliolumbar ligament mark the last lumbar vertebra independently of any count.^{1,32} If those differences hold at the boundary, the phenotype is decidable locally, and on this dataset it is. Separating T12 from L1 on shape alone gives an area under the curve of 0.990 and 97.3% accuracy over 1485 vertebrae. This was calculated using a logistic regression on five released per-level measures and six ratios between them. Each measure was divided by the patient’s median across their levels, so that size, which separates thoracic from lumbar trivially and says nothing at the junction, is removed. Folds are grouped by case and the curve is taken on pooled out-of-fold scores (`test_morphometric_separability.py` in the released code). Schinz et al. reach the same conclusion from the other side: on 1,242 whole-thoracolumbar CTs a shape-based labeling of the junction matched nerve morphology in every case, against 92.6–97.2% for counting- and rib-based rules.³³

II. ACQUISITION AND VALIDATION METHODS

II.A. Sources, and why a crosswalk was necessary

CTSpine1K’s 1,005 volumes and CTPelvic1K’s 1,184 come from four and seven collections of CT obtained for different indications. Only the Cancer Imaging Archive (TCIA) CT colonography collection (COLONOG)^{22,23} is common to both, and it is the largest in each: a subset of a prospective screening trial of asymptomatic patients aged fifty and over, acquired under one protocol. Every COLONOG patient was scanned prone and supine at a minimum, with 12 exceptions among the 825 patients the sources drew on. CTSpine1K states that one position was chosen at random by a released script; that script, since deleted but preserved in the repository’s history, writes every series holding twelve or more files to the same filename, so the volume retained is whichever series the directory walk reached last. No code was released with CTPelvic1K. Consistent with two independent choices, the sources annotated the same series for 342 of the 693 patients both annotated (49.4%), and the released records mix positions and reconstructions accordingly: 422 supine, 377 prone, three decubitus; 562

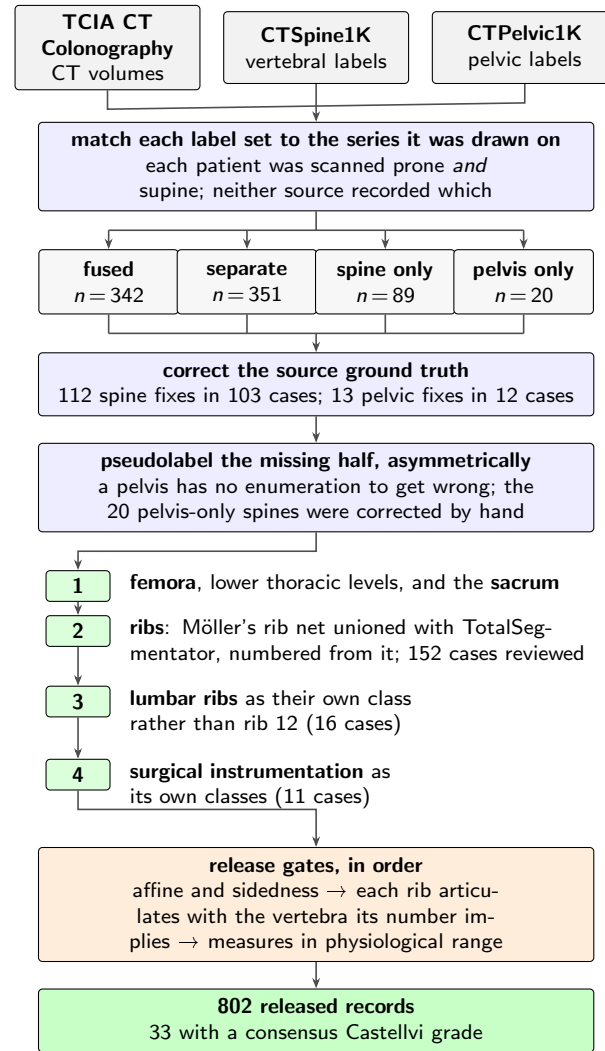


FIG. 1 How the dataset was built. n counts scans, or corrections, at each step.

at 1.0 mm, 232 at 1.25 mm and eight at other slice thicknesses.

Recovering the archived series each annotation was drawn on is nontrivial because the DICOM attributes that identify an acquisition, the patient, study and series instance identifiers and the patient position,³⁶ have no counterpart in the NIfTI-1 header³⁷ and are discarded on conversion.³⁸ CTSpine1K’s `Path.csv` names the TCIA directory each label was converted from, but it carries no series identifier, names no position for 217 of the 782 annotated records, and for seven indicates an acquisition other than the one whose bone the label overlaps best. CTPelvic1K released no imaging and encoded the mapping in its mask filenames, `dataset2.PatientID.SeriesNumber.undocumented.[tag.]mask_4label.nii.gz`. One of the 714 masks names a patient with no archived series; of the other 713, the first two fields resolve 702 to the series selected here by bone overlap, five to a `SeriesNumber` two series share, and six to one no series carries. The third field matches nothing, and the fourth, on 32 masks, mixes phenotype (`sacralization`

TABLE I Public CT collections at the lumbosacral junction. Black, class present; outlined, graded only to exclude it³⁴.

Collection	Scans	Count	L6	Sacrum	S1	Pelvis	Ribs	L. rib	Femora	Grade
CTSpine1K ²⁵	1,005									
CTPelvic1K ²⁴	1,184									
VerSe ¹⁰	374									
RibSeg v2 ³⁵	660									
TotalSegmentator ⁹	1,204									
CTSpinoPelvic1K	802									

7, semisacralization 2, hard_sacralization 1), difficulty (hard 8, veryhard 2), image condition (lowdose 6; metal, crop, IntestinalCalculus once each) and two unexplained tags (dqjoint 2, ydjoint 1) in one string.

The crosswalk (Fig. 1) therefore resolved every annotation, not only the 27.7% with no stated position, by thresholding each of the patient’s volumes at bone attenuation and keeping the series the mask overlaps most; where a source did name a series, 558 of the 565 CT-Spine1K records naming a position and 702 of the 713 CTPelvic1K masks naming an archived patient agree with it. The result is 342 records whose spine and pelvic labels land on the same series (*fused*), 351 whose labels land on different series of one patient (*separate*), 89 with a spine label only and 20 with a pelvic label only. Turning a patient between prone and supine alters lumbar alignment and the seating of the pelvis against it³⁹, so a mask drawn on one series does not transfer rigidly to the other, and while a patient’s two annotations lie on different scans no spinopelvic measurement can be taken across them.

II.B. Densification, and why it is asymmetric

A rigid cross-registration of each pelvic mask onto the patient’s other acquisition was attempted, and is released in the code base, but was found wanting. The current nnU-Net⁴⁰ framework has an `ignore` label that permitted training on pelvis-only, spine-only and fused masks together without degrading segmentation at the lumbosacral junction. A preliminary network was trained in five folds to pseudolabel the missing pelvises of every volume in the cohort, excluding those with fused masks, using out-of-fold inference so that no pelvis was pseudolabelled by a model that had seen it during training. This detail is important because it permitted comparing the accuracy of the model against the ground-truth radiologist-annotated pelvises: on the manual pelvic labels of each held-out fold the model reaches a Dice of 0.980 for the sacrum and 0.97 for each hip (Table II). The pelvis is thus radiologist-derived on the 362 fused and pelvis-only records and pseudolabeled on the other 440 (Fig. 1). Those 440 pseudolabelled pelvises are close to radiologist quality: a sacrum Dice of 0.980 ± 0.003 with no fold below 0.977, and hip Dice of 0.970 ± 0.016 and 0.971 ± 0.015 with no fold below 0.953, mean the model and the radiologist agree on 97–98% of the voxels of every pelvic bone on records the model never saw; the per-fold values are released with the data

(`results/pseudolabel_dice/`). The direction is asymmetric because a pelvis carries no enumeration to get wrong, whereas a pseudolabeled spine must commit to a count and, on a transitional vertebra, commits to the commoner reading; the 20 pelvis-only spines were corrected by hand. The pseudolabels also pass the release gates every structure passes (Sec. II.F), and the spinopelvic measures that depend on them agree with published supine values.

TABLE II Held-out fold Dice of the pelvic pseudolabeller against manual CTPelvic1K labels, mean $\pm$ SD over the five folds at each fold’s selected checkpoint (nnU-Net validation Dice, ignore regions excluded; `results/pseudolabel_dice/` in the repository).

Structure	Dice
Sacrum	0.980 ± 0.003
Left hip	0.970 ± 0.016
Right hip	0.971 ± 0.015

II.C. Ribs

No public rib annotation covers this cohort, and the two available automatic sources^{9,41} fail in complementary ways.

Möller’s binary rib network⁴¹ reports high accuracy, and its authors show that stump ribs can be classified from a partial field of view; that result concerns the morphological features computed on a rib once segmented. Here, though, the segmentation itself failed on ribs that were *partially outside* the field of view. A rib crossing the boundary of an abdominal acquisition was liable to be missed altogether rather than segmented to the edge. In a colonography cohort the lower ribs are routinely clipped, and those happen to be the ribs the count depends on.

TotalSegmentator⁹, by contrast, does segment clipped ribs and supplies the per-rib numbering that a binary network cannot. Its characteristic failure is at the other end of the bone, where the most medial portion of the rib (roughly the last tenth to sixth approaching the costovertebral joint) is frequently absent. A rib truncated at its medial end never reaches the vertebra it articulates with, so the rib–vertebra incidence check (the quality-control gate on the rib class labels, which matches each rib to the vertebra its number implies) has nothing to test.

Given these complementary gaps, the two approaches were combined (Fig. 1), TotalSegmentator supplying the numbering and the medial-clipped cases, Möller the detection where its output is more complete.

The result of their union is a pseudolabel of all ribs in the volume. An extensive quality control pipeline was

used to triage possible errors rather than exhaustively reviewing all the pseudolabels (Table III). Two label-based checks gated the triage: a rib bone carrying two identifiers, and a rib that never reaches the spine. They selected 152 records; 148 were reviewed and corrected, and four were not and are identified in the release, which is where the residue in the first two rows sits. Reviewers were medical students working through OpenSpineConsortium⁴², trained against a written labeling protocol, with every correction attributed to its annotator, and a save was refused until the review checks passed. Ribs are named for the vertebra they articulate with, so after review the remaining whole-side shifts were renumbered against the vertebrae by rule; the release-stage incidence is therefore a consistency check on that rule, and the two earlier stages are the independent measure of the pseudolabel and of the reviewers.

TABLE III Rib checks on the raw pseudolabel and after the student corrections. The first two gated the review; the rest were measured on every record at both stages.

Check	Before	After
One rib bone, two numbers (records / bones)	151 / 584	3 / 2
Rib detached from the spine (records)	38	
One number in several pieces (records / pieces)	463 / 1,146	330 / 378
Rib offset from its vertebra (ribs / records)	170 / 24	109 / 161
Rib with no vertebra within reach (ribs)	38	12
Rib on a lumbar body, numbered thoracic (ribs)	21	25

II.D. Label scheme and counting anchors

The scheme is VerSe-native. Vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12 = 8–19, L1–L6 = 20–25, sacrum 26, coccyx 27, T13 = 28), and every non-VerSe structure takes a fixed identifier above that range, hips (30–31), femora (32–33), thirteen ribs per side (34–46 left, 47–59 right), lumbar ribs (60–61) and surgical hardware (62–68). Identifier 29 is retired and unused, for the reason given below. Rib 13 is the true rib of a T13 and is empty in this release, so a thirteenth thoracic vertebra and a lumbar rib are recorded as the distinct phenotypes they are.

Two anchors give every record the same positional frame, and Fig. 2 shows both, the lowest rib-bearing vertebra above and the **sacrum** below. The sacrum’s superior end-plate is the landmark sacral slope and pelvic incidence are measured from, and with both brackets present L5 and L6 are distinguishable where a spine-limited view alone would not be.

A separate S1 label was withdrawn, and why. Earlier releases carved the cranial portion of the sacrum as its own S1 class, on an automatic estimate of the S1–S2 boundary. Measured against the anatomy on all 802 records, that estimate is unreliable: S1’s craniocaudal extent is a median 57.6 mm where a first sacral segment is near 30, only 26% of records fall in a defensible range, and in 222 the plane takes more than half the sacrum. The failures are not merely low cuts. Where the fit degenerates the plane comes out nearly parallel to the sacral

long axis and shaves the ventral cortex, so the retained sacrum is left without its anterior wall — wrong for every use of the label, not only for the end-plate. Rendering one such record is what showed this; no summary statistic did.

The consequence was confined to the parameters fitted to that surface. Splitting the cohort on whether the carve was anatomically sound separates them cleanly: the 580 sound records give a pelvic incidence of 52.7° and a pelvic tilt of 19.6°, against Hasegawa’s supine-CT 53.4 and 19.2, while the 221 degenerate ones give 68.5 and 33.6, and the end-plate gate described in Sec. II.E fires on 96% of the second group and 4% of the first. The measurement was never at fault; the surface it was given was. So S1 is merged back into the sacrum, restoring the bone the source annotation actually delineated, and the end-plate is fitted to the whole sacrum instead.

VerSe¹⁰ carries L6, whose identifier this release keeps, so its L6 *is* VerSe’s.

A rib on a lumbar body receives its own class,

After and this is the one class here with no published counterpart. A lumbar rib and a stump rib are the same object under two counts, so a scheme that numbers every rib 12 calling it rib 12, which asserts that the vertebra beneath it is thoracic, the very question at issue, or discarding it. TotalSegmentator carries ribs 1–12 per side and no such class, and VerSe’s T13 is a vertebra rather than a rib. This scheme records both: VerSe’s T13 (28) with its rib pair (46, 59) for a thoracic-type thirteenth vertebra, and 60/61 for a lumbar-type body bearing a rudimentary rib. No record here carries a T13; the sixteen lumbar-rib cases were read as lumbar-type bodies, and where a border vertebra could not be settled by its readers the label stands as read.

Figure 2 shows why the interval is the primitive and the label is derived. Panels (b) and (c) present the same upper border, a rib pair of about 40 mm beneath a full pair, above four rib-free vertebrae; the source readers named it a lumbar rib on L1 in (b) and stump twelfth ribs on T12 in (c). Nothing in the rib decided that. Across the release the sixteen ribs labeled lumbar are a median 45 mm and 0.32 of the rib above them, and the 98 labeled stump twelfth ribs 38 mm and 0.28. What decided it was the lower border, no transitional vertebra in (b) and a fifth lumbar segment fused to the sacrum in (c) (Castellvi IIb), together with the convention of five lumbar vertebrae, which leaves one place for the anomaly in each. The count cannot separate the two; the rib length and the lumbosacral junction, both labeled here, can.

Sixteen records carry a lumbar rib, thirteen bilateral and three unilateral, all on the right, too few to speak to side preference.

Eighteen records carry an L6. Seventeen of them carry a consensus Castellvi grade; the eighteenth was not graded. The source transitional label (pelvic where CT-Pelvic1K flagged one, otherwise the lumbar count) reads lumbarization in fourteen, sacralization in two and semi-

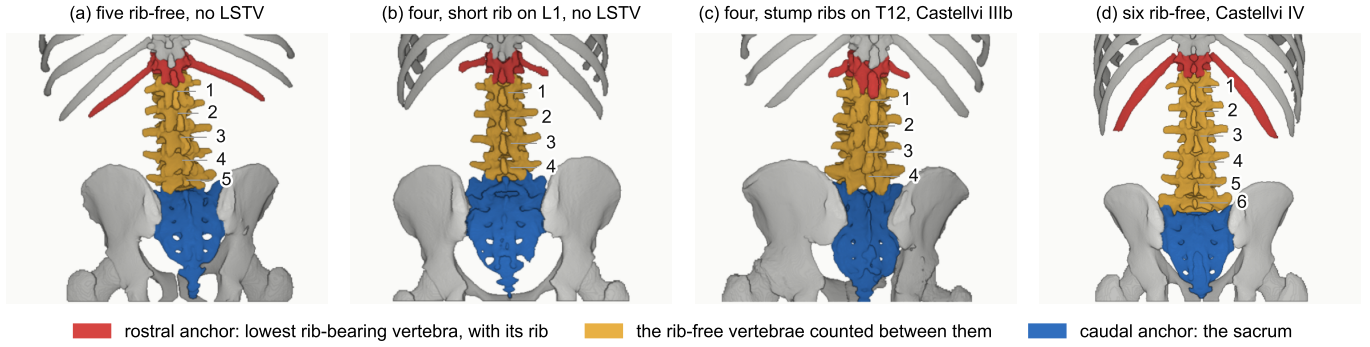


FIG. 2 The two anchors, aligned on the sacrum. Red, lowest rib-bearing vertebra and its rib; blue, sacrum; yellow, rib-free vertebrae between. Cases 0704, 0428, 0094, 0005, chosen by rule. (b) and (c) share one upper border, a short rib pair under a full pair; (b) has no LSTV; (c) is Castellvi IIIb, its fifth lumbar segment fused to the sacrum.

sacralization in one, and is unremarkable in the ungraded record. A spine can carry six lumbar-type vertebrae and still be read as a sacralization, depending on where the reader started the count.

Both counts, and the manifest fields that report them (`has_l6`, `has_lumbar_rib`, `n_lumbar_labels`, `lumbar_rib_side`), are computed from the released label volumes by counting identifiers 25 and 60–61.

II.E. Computational tools

Access. Every step is performed by open-source code archived with the dataset, together with the reference loader, the label scheme and the quality-control scripts that generated the tables in this manuscript.

Versions and parameters. Segmentation of femora, the sacral sub-division and the per-level rib numbering used TotalSegmentator⁹ (total task, release 2.x) on a single graphics processing unit. The binary rib network is Möller’s⁴¹ released weights, applied through the nnU-Net v2 framework⁴⁰. The pelvic completion for records whose pelvis was absent used a five-fold nnU-Net v2 ensemble applied out-of-fold, so that no record is completed by a model that has seen it. The five fold checkpoints, with their nnU-Net plans and dataset descriptor, are released alongside the data (<https://huggingface.co/OpenSpineConsortium/spinopelvic-seg-checkpoints>, trained with <https://github.com/Gregory-Schwing-MD-PhD/spinesurg-ct-nnnet> at commit 50b209f). Image handling throughout used NiBabel and SciPy; no image is resampled or reoriented when a label is written, so every label shares its image’s grid and affine exactly.

Measurement. Dimensions and spinopelvic parameters come from code released with the dataset. A sacral plate whose fitted normal lies more than 60° off the cranial axis is rejected and the case reported missing, since such a fit has found the near-vertical anterior face of the promontory, not the end-plate. Gating on the geometry rather than on the value leaves unusual patients in. OpenSpineToolkit, a unit-tested open-source package is released separately (<https://github.com/OpenSpineConsortium/>

[OpenSpineToolkit](#)) as a reusable implementation measuring in each vertebra’s own frame; no open-source implementation computing these parameters from CT segmentations was found elsewhere.

Generative artificial intelligence. A large language model (Claude, Anthropic) assisted with writing the analysis and figure code and manuscript refinement after initial drafting. It was not used to generate, annotate or interpret imaging data; every number reported here is computed by the released code from the released volumes. All content was independently reviewed, verified, and interpreted by the authors.

II.F. Validation

Validation has three layers. Geometry is checked first, because a measurement taken on a corpus with a geometric fault still returns a plausible-looking number.

Geometric invariants. Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for sided structures falling on the correct sides, with the left–right axis read from the affine rather than assumed.

The check compares *each sided pair separately*. Testing the ribs alone passed all 802 records, four of which carried a `left_hip` label on the patient’s right side; pooling every sided structure into one test still missed one of those four records, because a large correctly-sided structure masks a smaller transposed one. A gate that passes everything is not evidence of a clean corpus until it has been shown capable of failing.

Rib–vertebra incidence. Each rib is matched to its nearest vertebra and compared against the vertebra its number implies. Of 11,548 ribs, 5,788 are not evaluable because the expected vertebra lies outside the field of view and 11 have no vertebra within the anchor distance; of the 5,749 evaluable, 5,747 match, two (0.035%) are offset by one level, and no case is misnumbered. The denominator is stated because quoting the two offsets against all 11,548 ribs would halve the rate by counting ribs the check never examined. Both offsets are field-of-view truncations, one at +1 level and one at −1.

The same check carries an invariant that reads as a null result. Of the 5,749 evaluable ribs, *zero* are assigned to a lumbar vertebra, not because no lumbar ribs exist, since 16 records carry one, but because such a rib takes class 60 or 61 and never enters the numbered set the test examines. A numbered rib landing on a lumbar vertebra would be a counting error of exactly the kind this dataset exists to expose.

Agreement with published values. Pelvic incidence is the strongest of these checks because it is a morphological property rather than a posture; it agrees to within half a degree with the automated supine CT series of Veilleux *et al.*⁴³ and sits below the standing figure, the direction reported for subjects imaged both ways.⁴⁴ Sacral slope and pelvic tilt are postural, and read three degrees below and four above Veilleux’s asymptomatic subjects; a second supine CT series reports 53.4, 34.1 and 19.2°, within a degree of this cohort on all three.³⁹ Recumbency runs the wrong way to explain the gap, raising slope rather than lowering it, so the residual tracks cohort and not construction.

II.G. Castellvi typing

Every record whose vertebral count departs from five carries a Castellvi grade² in the released manifest, 33 cases typed I–IV with the *a/b* unilateral–bilateral qualifier. The grade and the count are *different axes*. Grade IIb occurs here at rib-free counts of four, five and six alike, and seven of the 33 graded cases carry a normal count of five.

Two radiology resident physicians graded every case independently and resolved their six disagreements by consensus; both reads and the consensus are in the manifest. The distinction the readers disagreed on, type II (articulation) against type III (bony fusion)¹, is the one that matters clinically, and four of the six disagreements were III against II.

II.H. Surgical hardware, and why a threshold is not enough

Metal is trivial to detect and hard to interpret, and the difference matters here because **an iatrogenic fusion is indistinguishable from a congenital one to a distance measurement**. A cage-bridged interspace reads as “no gap” exactly as a fused transitional vertebra does.

II.H.0.a. Detection over-calls by a factor of five. A 2500 HU threshold inside a shell around the labeled skeleton flags 52 of the 802 records. All 52 were reviewed manually, confirming 11 and rejecting 41 (79%) as contrast, calcification and reconstruction artefact. Attenuation does not separate the two groups (both saturate); site and volume do, every confirmed implant at 2,586 mm³ or above and every rejection at 1,768 mm³ or below.

II.H.0.b. The subtype block had to be extended. Identifiers 62–65 name spinal instrumentation (generic, cage, screw-and-rod, plate) and cannot describe what this cohort holds, so **66 arthroplasty**, **67 sacroiliac screw**

and **68 osteosynthesis** were added. Osteosynthesis holds parts of one bone together where arthroplasty replaces a joint, and a shape rule calls a femoral stem a rod. The confirmed cases are eight arthroplasties, one osteosynthesis, one sacroiliac fixation and one pair of interbody cages (Fig. 3).

II.H.0.c. Metal outranks bone, and that is a subtraction. In every confirmed case the implant was already inside a bone label, so naming it reclaims 1,538,852 voxels rather than adding them. In eight of these cases the femoral head that pelvic incidence and tilt are measured from is an implant.

III. DATA FORMAT AND USAGE NOTES

Each record is a gzipped NIFTI image/label pair sharing an identical 4 × 4 affine, canonicalized to PIR, requiring no resampling. Masks are NIFTI rather than DICOM, which is what volumetric segmentation tooling expects.

Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the data as `splits_5fold.json`. The five validation folds partition all 802 records, so the release provides cross-validation and *not* a held-out test set; users reporting a single held-out number should carve one out and state which records it holds.

Stratification is on the transitional subtype rather than a binary LSTV flag. Across all 802 records the source transitional label reads lumbarization in 14, sacralization in 17 and semi-sacralization in two; separately, 18 records carry a sixth lumbar-type vertebra, nine carry only four lumbar labels, and 33 carry a consensus Castellvi grade. A record called sacralization by its four lumbar labels enters the same stratum as one called sacralization by the source label, so the two rare strata hold 15 records each and every validation fold carries three of each. That is the most stratification can guarantee, and it is why a fold-level metric on the rare classes carries an error bar far wider than its own decimal places.

The archive of record is Zenodo (<https://doi.org/10.5281/zenodo.22139642>). Code and documentation are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror, not the archive.

IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records: 393 recorded female and 345 male, 11 carrying DICOM’s *other* value and 53 no sex field, with age present for 709 (median 59, range 50–89). All 802 are counted in totals; the eleven outside the female/male dichotomy are excluded from sex-stratified measures.

Count-free measures. Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal, and its lower mode, below 0.33 of the rib above, marks a stump twelfth rib in 98 of the 789 records

TABLE IV Spinopelvic measures, mean $\pm$ SD over the 756 of 802 records passing the geometric gates, against two supine CT series that report means.^{39,43} Standing values are 55.0, 41.0, 13.0 and 60.0°.⁴⁵ The end-plate is fitted to the whole sacrum; the gate rejects 14 records where the previous carved-S1 fit lost 239. Pelvic incidence and tilt are right-skewed (0.25 and 0.33), so their medians—51.5 and 15.0°—sit below these means and the modal peak in Fig. 4 sits below both.

Measure	This dataset (supine, $n = 756$)	Veilleux ($n = 200$)	Hasegawa ($n = 24$)
Pelvic incidence	52.1 $\pm$ 11.1°	52.1	53.4
Sacral slope	36.6 $\pm$ 10.6°	36.5	34.1
Pelvic tilt	15.5 $\pm$ 7.9°	15.6	19.2
Lumbar lordosis	51.7 $\pm$ 13.3°	—	—
PI – LL mismatch	0.3 $\pm$ 10.3°	—	—

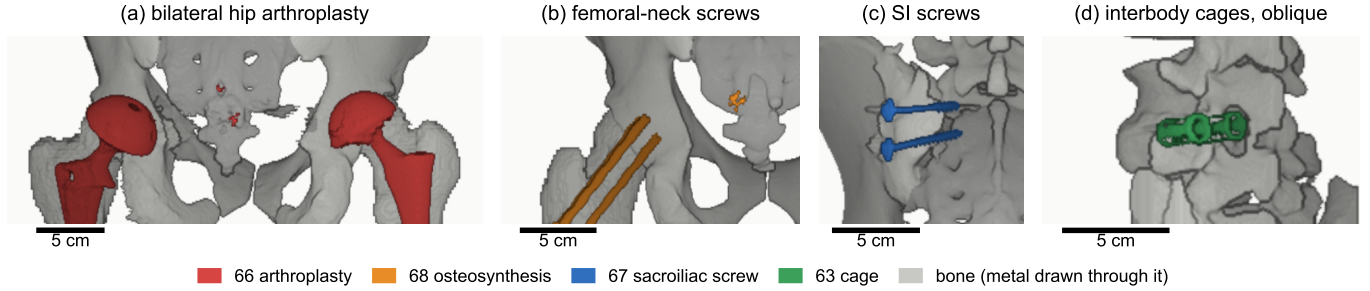


FIG. 3 Instrumentation in the release, one exemplar each, drawn through bone. (a) Hip arthroplasty, 66, $n = 8$. (b) Femoral-neck screws, 68, $n = 1$. (c) Sacroiliac screws, 67, $n = 1$. (d) Interbody cages, 63, $n = 1$. Bar, 5 cm.

TABLE V Data types and metadata, and the records populating each, from the released manifest.

Field	Records	Available
Identifiers and paths, configuration, match type, image/label alignment check	802	100.0%
Patient position, tube potential, section thickness, kernel	802	100.0%
Transitional label and class, annotation origin, instrumentation flags	802	100.0%
Spine series identifier; bone-fraction check	782	97.5%
Scanner manufacturer and model	768	95.8%
Sex	749	93.4%
Age	709	88.4%
Pelvic series identifier	713	88.9%
Castellvi grade (consensus of two readers)	33	4.1%

with a measurable pair (12.4%, against 12.6% on whole-spine CT⁸); 16 records carry a lumbar rib (2.0%, against 1.5%). The co-occurrence reported there is testable here only where the junction was read, and only within those 789: they hold 14 of the 17 source-labeled sacralizations, and stump ribs accompany four of those 14 and 94 of the other 775 (odds ratio 2.9, $p = 0.08$, Fisher’s exact test). No lumbarization carries a lumbar rib.

Spinopelvic measures (Fig. 4). Pelvic incidence sits on the standing reference, as a morphological property should; slope and tilt sit below and above the published CT values, which Table IV places against two series.

Opportunistic measures. Every scan carries a vertebral bone density measurement at no additional dose.⁴⁶ L1 trabecular attenuation is reported at the published standard site.

V. POTENTIAL APPLICATIONS AND LIMITATIONS

The principal use is the *spinopelvic interface*. Spine and pelvis carry radiologist-sourced annotation on one coordinate frame in 802 records, making sacral morphology measurable against the lumbar column rather than in isolation, among them the sacral endplate and its slope, the alar corridors constraining S2-alar-iliac and il-

iac screw trajectories, and the L5–S1 geometry a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and femoral heads, so it is reproducible from the labels themselves.

The transitional layer serves that use rather than competing with it, since no trajectory can be planned across a junction whose segments cannot be named.

A proxy for the preoperative view. No public preoperative lumbar CT cohort exists that we know of, so this release is the closest available stand-in for the view a lumbar case is planned on: it holds the T12-to-S1 span with the lowest ribs and the pelvis, 377 of its 802 records were acquired prone, the position of posterior lumbar surgery, and it carries classes for the anomalies that make level identification fail. It is a screening population at colonography dose, not a surgical series, so it supports method development for level identification and instrumentation geometry rather than validation of a planning decision.

V.A. Three uses the released measurements support

Naming a level from its own geometry. Wrong-level spine surgery runs at roughly one in 3,110 spinal procedures, and among the predominant contributing factors is the transitional anatomy this cohort was assembled around.^{47,48} Per-level body height, canal width, endplate

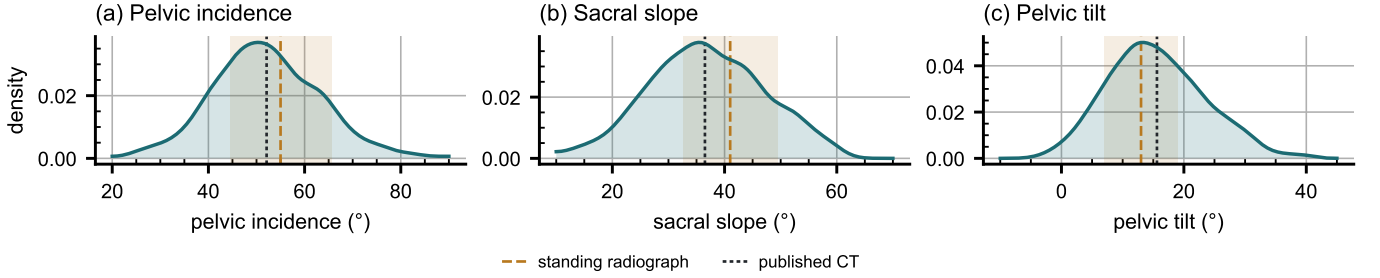


FIG. 4 Spinopelvic measures against standing reference bands (mean $\pm$ SD)⁴⁵.

width, transverse-process span and wedge ratio are released across all 802 records, and because the labels were assigned against the twelfth rib rather than by enumeration, a model trained on them is not learning the convention that fails.

Reference morphometry with its spread. The values a surgeon consults for endplate and canal dimensions and pedicle width come from cadaveric series of ten to thirty specimens^{29–31} redrawn in the textbooks as one curve per level⁴⁹, and a standard error describes the mean, not the spread of individuals. The same measures are given here from 802 records with the distribution attached (Fig. 5), reproducing the classical pedicle widening into the lower lumbar spine. Both layers of that figure are population intervals rather than confidence intervals, and that distinction is what makes their widths comparable: a confidence interval bounds a parameter, a population interval bounds individuals,⁵⁰ and at $n = 12$ the two differ by $\sqrt{12}$.

Patient-specific models. Patient-specific finite-element modeling for alignment planning and implant design⁵¹ needs spine, pelvis and femoral heads in one frame, which is this release’s construction, and 351 patients give a within-patient postural change to check a prediction against.

Limitations: coverage and cohort. Thoracic ground truth is field-of-view limited (Fig. 6) and does not reach T1. Postural angles are supine; pelvic incidence is not postural, but it is modality-sensitive, reading below radiography for the same subjects.⁴⁴ End-plate and pedicle width are withheld at L5, where the transverse processes arise in front of the canal wall the body is cut at. Those two and the spinopelvic angles, which rest on the sacral end-plate, are the provisional part of this release; the planned revision takes body and sacral plate from a substructure label, and the version DOI carries it. The cohort is one protocol of a colorectal screening population aged 50 and over, so its distributions should not be read as a surgical one.

Label strength varies by structure, and the release does not average over that. Vertebral labels derive from radiologist-supervised source annotations, corrected where those sources were wrong; pelvic labels on records that lacked one are pseudolabeled. The rib layer is a pseudolabel whose review was triaged, so its guarantee is the absence of the failure modes that rule detects. Transitional labels come from two independent sources and

are not uniformly adjudicated, which is why the measures reported here are count-free; the Castellvi grades are a consensus of two radiology resident physicians. The sacrum is the source annotation’s own outer boundary throughout; no sub-division of it is asserted, the separate S1 label of earlier releases having been withdrawn as unreliable.

Absent structures, and no classifier at the lower border. Sacrum, hips and femora are present in all 802 records; one lacks T12, outside the field of view. Nine carry no L5 identifier because their source annotation counted four lumbar vertebrae; all nine are Castellvi IIIb, the fused segment is delineated with the sacrum as the caudal anchor, and the manifest names them. No shape-based classifier is attempted at the lumbosacral junction: 15 records in each rare stratum leave three per fold. The 0.990 area under the curve above is the thoracolumbar boundary, not this one.

VI. DISCUSSION

Comparison with related material. VerSe^{10,34} comes closest in intent to this dataset, but carries no sacral mask for the structure a Castellvi grade describes, which this release segments as L6 or as sacrum, and has neither the pelvis and femora a spinopelvic measurement needs nor prone acquisitions. What this release adds over TotalSegmentator⁹ is the classes for anatomical anomalies, not necessarily more anatomy per scan.

Future directions. Applying the released weights to VerSe would add the sacral anchor and the anomaly classes to a bone-kernel collection. A Castellvi read of the whole cohort would test the co-occurrence of rib and lumbosacral anomalies at full power and supply the labels a lumbosacral classifier needs. A class label from substructure segmentation¹¹ would measure the body without the canal cut that withholds L5 here.

VII. CONCLUSION

CTSpinoPelvic1K places the radiologist-derived spine, the pelvis, radiologist-derived where the sources annotated the same series and pseudolabeled elsewhere, and pseudolabeled per-level ribs and femora on one validated coordinate frame in 802 records, and gives the anatomical anomalies that make lumbar numbering ambiguous classes of their own. Because both counting anchors are

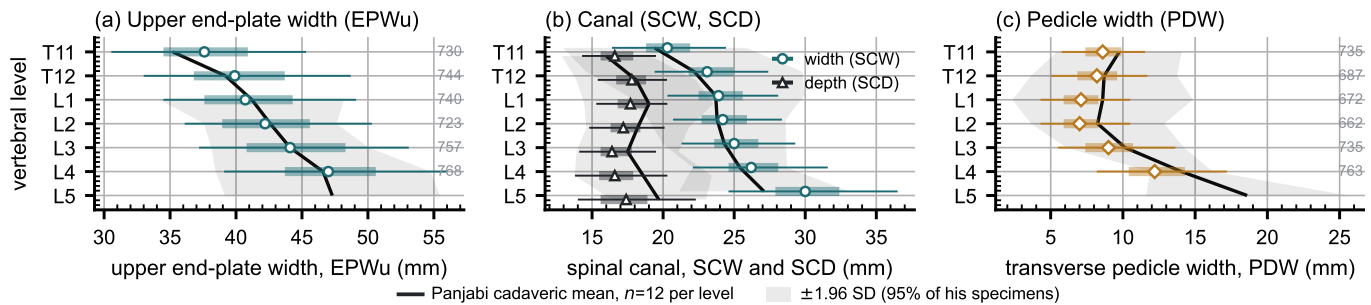


FIG. 5 Morphometry by level, T11–L5: median, interquartile range, 5th–95th percentile, n at right. (a) End-plate width EPWu. (b) Canal width SCW and depth SCD. (c) Pedicle width PDW, both sides averaged. L5 is withheld in (a) and (c); see text. Reference is Panjabi's cadaveric series^{29,30}, keyed in (a), its SD recovered from the reported SEM; T11–T12 EPWu is digitised from his Figure 4A.

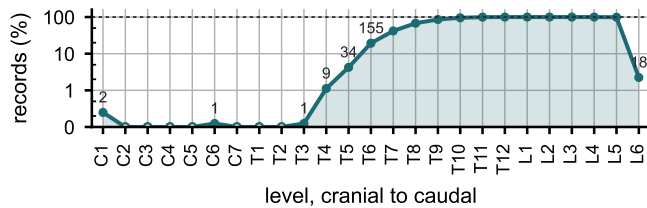


FIG. 6 Records carrying each vertebral level, cranial to caudal. Counts per identifier in Table S1.

explicit, a level can be identified from a field of view that cannot support the conventional count downward from C2.

DATA AVAILABILITY

The release described here is v10, permanently archived at <https://doi.org/10.5281/zenodo.22642578>; the concept DOI <https://doi.org/10.5281/zenodo.22139642> resolves to the current version. The archive carries the loader, the label scheme and the quality-control tables under an open-source licence. The repository additionally carries the extraction and figure code, the reference table with a citation on every row, the values plotted in Fig. 5, and a recipe for running them in order. Code: <https://github.com/OpenSpineConsortium/CTSpinoPelvic1K> (article materials under `paper/mpda/`); convenience mirror: <https://huggingface.co/datasets/OpenSpineConsortium/CTSpinoPelvic1K>; measurement toolkit: <https://github.com/OpenSpineConsortium/OpenSpineToolkit>.

ACKNOWLEDGEMENTS

Acknowledgements are given on the title page, per the journal's double-anonymized review policy. This work received no funding.

CONFLICT OF INTEREST

The authors have no relevant conflicts of interest to disclose.

ETHICS

This work uses publicly available, de-identified imaging and annotations derived from it. The authors' institutional review board determined that it is not human participant research under 45 CFR 46 and requires no oversight.

a)Electronic mail: gregory.schwing@med.wayne.edu

- ¹G. Konin and D. Walz, Lumbar transitional vertebrae: classification, imaging findings, and clinical relevance, *AJNR Am. J. Neuroradiol.* **31**, 1778–1786 (2010), doi:10.3174/ajnr.A2036.
- ²A. E. Castellvi, L. A. Goldstein, and D. P. Chan, Lumbar transitional vertebrae and their relationship with lumbar extradural defects, *Spine (Phila. Pa. 1976)* **9**, 493–495 (1984), doi:10.1097/00007632-198407000-00014.
- ³J. A. Carrino, P. D. Campbell, D. C. Lin, W. B. Morrison, M. E. Schweitzer, A. E. Flanders, J. Eng, and A. R. Vaccaro, Effect of spinal segment variants on numbering vertebral levels at lumbar MR imaging, *Radiology* **259**, 196–202 (2011).
- ⁴N. A. Farshad-Amacker, B. Lurie, R. J. Herzog, and M. Farshad, Merits of different anatomical landmarks for correct numbering of the lumbar vertebrae in lumbosacral transitional anomalies, *Eur. Spine J.* **24**, 600–608 (2015).
- ⁵D. Tureli, G. Ekinici, and F. Baltacioglu, Is any landmark reliable in vertebral enumeration? A study of 3.0-Tesla lumbar MRI comparing skeletal, neural, and vascular markers, *Clin. Imaging* **38**, 792–796 (2014).
- ⁶J. Lian, N. Levine, and W. Cho, A review of lumbosacral transitional vertebrae and associated vertebral numeration, *Eur. Spine J.* **27**, 995–1004 (2018), doi:10.1007/s00586-018-5554-8.
- ⁷J. Hanhivaara, J. H. Määtä, J. Karppinen, J. Niinimäki, and M. T. Nevalainen, The association of lumbosacral transitional vertebrae with low back pain and lumbar degenerative findings in MRI: a large cohort study, *Spine* **47**, 153–162 (2022).
- ⁸S. Nagata et al., Lumbar transitional vertebra on whole-spine CT: prevalence and association with rib abnormalities, *Skeletal Radiol.* **54**, 2169–2177 (2025), doi:10.1007/s00256-025-04952-z.
- ⁹J. Wasserthal et al., TotalSegmentator: robust segmentation of 104 anatomic structures in CT images, *Radiol. Artif. Intell.* **5**, e230024 (2023), doi:10.1148/ryai.230024.
- ¹⁰A. Sekuboyina et al., VerSe: a vertebrae labelling and segmentation benchmark for multi-detector CT images, *Med. Image Anal.* **73**, 102166 (2021), doi:10.1016/j.media.2021.102166.
- ¹¹H. Möller et al., SPINEPS—automatic whole spine segmentation of T2-weighted MR images using a two-phase approach to multi-class semantic and instance segmentation, *Eur. Radiol.* **35**, 1178–1189 (2025), doi:10.1007/s00330-024-11155-y.
- ¹²H. Möller et al., VERIDAH: Solving Enumeration Anomaly Aware Vertebra Labeling across Imaging Sequences, *arXiv*, 2026, arXiv:2601.14066. doi:10.48550/ARXIV.2601.14066.
- ¹³Y. Otake et al., Automatic localization of vertebral levels in x-ray fluoroscopy using 3D-2D registration: a tool to reduce wrong-site surgery, *Phys. Med. Biol.* **57**, 5485–5508 (2012), doi:10.1088/0031-9155/57/17/5485.

- ¹⁴S.-f. L. Lo et al., Automatic Localization of Target Vertebrae in Spine Surgery: Clinical Evaluation of the LevelCheck Registration Algorithm, *Spine (Phila. Pa. 1976)* **40**, E476–E483 (2015), doi:10.1097/BRS.0000000000000814.
- ¹⁵T. De Silva et al., Utility of the LevelCheck Algorithm for Decision Support in Vertebral Localization, *Spine (Phila. Pa. 1976)* **41**, E1249–E1256 (2016), doi:10.1097/BRS.0000000000001589.
- ¹⁶T. A. Hutchins et al., ACR Appropriateness Criteria Low Back Pain: 2021 Update, *J. Am. Coll. Radiol.* **18**, S361–S379 (2021), doi:10.1016/j.jacr.2021.08.002.
- ¹⁷D. S. Kreiner et al., An evidence-based clinical guideline for the diagnosis and treatment of degenerative lumbar spinal stenosis (update), *Spine J.* **13**, 734–743 (2013), doi:10.1016/j.spinee.2012.11.059.
- ¹⁸American College of Radiology, ACR–ASNR–ASSR–SPR practice parameter for the performance of computed tomography (CT) of the spine, revised 2022 (Resolution 23), American College of Radiology, Reston, VA. <https://www.asnr.org/wp-content/uploads/2019/06/CT-Spine-1.pdf>, 2022.
- ¹⁹American College of Radiology, ACR–ASNR–SABI–SSR practice parameter for the performance of magnetic resonance imaging (MRI) of the adult spine, revised 2023 (Resolution 6), American College of Radiology, Reston, VA. <https://www.asnr.org/wp-content/uploads/2019/06/MR-Adult-Spine.pdf>, 2023.
- ²⁰S. Sixta et al., Screening for thoracolumbar spinal injuries in blunt trauma: an Eastern Association for the Surgery of Trauma practice management guideline, *J. Trauma Acute Care Surg.* **73**, S326–S332 (2012), doi:10.1097/TA.0b013e31827559b8.
- ²¹American College of Surgeons Committee on Trauma, ACS TQIP best practices guidelines in imaging, American College of Surgeons, Chicago. https://www.facs.org/media/oxdjw5zj/imaging_guidelines.pdf, 2018.
- ²²K. Smith et al., Data from CT COLONOGRAPHY (ACRIN 6664), The Cancer Imaging Archive, version 2 [dataset], 2015, doi:10.7937/K9/TCIA.2015.NWTESAY1.
- ²³K. Clark et al., The Cancer Imaging Archive (TCIA): maintaining and operating a public information repository, *J. Digit. Imaging* **26**, 1045–1057 (2013), doi:10.1007/s10278-013-9622-7.
- ²⁴P. Liu et al., Deep learning to segment pelvic bones: large-scale CT datasets and baseline models, *Int. J. Comput. Assist. Radiol. Surg.* **16**, 749–756 (2021), doi:10.1007/s11548-021-02363-8.
- ²⁵Y. Deng et al., CTSpine1K: a large-scale dataset for spinal vertebrae segmentation in computed tomography, *Mach. Learn. Biomed. Imaging* **3**, 824–832 (2025), doi:10.59275/j.melba.2025-gf84.
- ²⁶A. M. Du Plessis, L. M. Greyling, and B. J. Page, Differentiation and classification of thoracolumbar transitional vertebrae, *J. Anat.* **232**, 850–856 (2018), doi:10.1111/joa.12781.
- ²⁷A. M. Poolman, Q. Wessels, A. V. Schoor, and N. Keough, Thoracolumbar transitional vertebrae: Quantitative differentiation and associated numeric variation in the vertebral column using skeletal remains, *J. Anat.* **243**, 311–318 (2023), doi:10.1111/joa.13865.
- ²⁸M. R. Zindrick et al., Analysis of the morphometric characteristics of the thoracic and lumbar pedicles, *Spine (Phila. Pa. 1976)* **12**, 160–166 (1987), doi:10.1097/00007632-198703000-00012.
- ²⁹M. M. Panjabi et al., Thoracic human vertebrae: quantitative three-dimensional anatomy, *Spine (Phila. Pa. 1976)* **16**, 888–901 (1991), doi:10.1097/00007632-199108000-00006.
- ³⁰M. M. Panjabi et al., Human lumbar vertebrae: quantitative three-dimensional anatomy, *Spine (Phila. Pa. 1976)* **17**, 299–306 (1992), doi:10.1097/00007632-199203000-00010.
- ³¹J. L. Berry, J. M. Moran, W. S. Berg, and A. D. Steffee, A morphometric study of human lumbar and selected thoracic vertebrae, *Spine (Phila. Pa. 1976)* **12**, 362–367 (1987), doi:10.1097/00007632-198705000-00010.
- ³²R. J. Hughes and A. Saifuddin, Numbering of lumbosacral transitional vertebrae on MRI: role of the iliolumbar ligaments, *AJR Am. J. Roentgenol.* **187**, W59–W65 (2006), doi:10.2214/AJR.05.0415.
- ³³D. Schinz et al., A vertebral shape-based classification for labeling the thoracolumbar junction on CT imaging, *Eur. Spine J.* **35**, 2195–2203 (2026), doi:10.1007/s00586-025-09364-0.
- ³⁴H. Liebl et al., A computed tomography vertebral segmentation dataset with anatomical variations and multi-vendor scanner data, *Sci. Data* **8**, 284 (2021), doi:10.1038/s41597-021-01060-0.
- ³⁵L. Jin et al., RibSeg v2: a large-scale benchmark for rib labeling and anatomical centerline extraction, *IEEE Trans. Med. Imaging* **43**, 570–581 (2024), doi:10.1109/TMI.2023.3313627.
- ³⁶National Electrical Manufacturers Association, Digital Imaging and Communications in Medicine (DICOM) Standard, Part 3: Information Object Definitions, Technical Report PS3.3, NEMA, Rosslyn, VA, 2024, Attributes (0010,0020), (0020,000D), (0020,000E), (0018,5100).
- ³⁷R. W. Cox, J. Ashburner, H. Breman, K. Fissell, C. Haselgrove, C. J. Holmes, J. L. Lancaster, D. E. Rex, S. M. Smith, J. B. Woodward, and S. C. Strother, A (sort of) new image data format standard: NIFTI-1, *NeuroImage* **22**, e1440 (2004), 10th Annual Meeting of the Organization for Human Brain Mapping, Budapest.
- ³⁸X. Li, P. S. Morgan, J. Ashburner, J. Smith, and C. Rorden, The first step for neuroimaging data analysis: DICOM to NIFTI conversion, *J. Neurosci. Methods* **264**, 47–56 (2016).
- ³⁹K. Hasegawa, M. Okamoto, S. Hatsushikano, G. Caseiro, and K. Watanabe, Difference in whole spinal alignment between supine and standing positions in patients with adult spinal deformity, *BMC Musculoskelet. Disord.* **19**, 437 (2018).
- ⁴⁰F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation, *Nat. Methods* **18**, 203–211 (2021), doi:10.1038/s41592-020-01008-z.
- ⁴¹H. Möller et al., Automated thoracolumbar stump rib detection and analysis in a large CT cohort, *arXiv*, 2025, arXiv:2505.05004. doi:10.48550/arXiv.2505.05004.
- ⁴²A. Schehr, J. Kim, and G. Schwing, OpenSpineConsortium: an open-source framework for medical student engagement in computational spine imaging research, *Cureus* (2026), doi:10.7759/cureus.112661.
- ⁴³N. J. Veilleux, N. V. Kalore, J. A. Vossen, and J. S. Wayne, Automatic characterization of pelvic and sacral measures from 200 subjects, *J. Bone Joint Surg. Am.* **102**, e130 (2020), doi:10.2106/JBJS.20.00343.
- ⁴⁴C. M. Lee and R. W. Liu, Comparison of pelvic incidence measurement using lateral x-ray, standard CT versus CT with 3D reconstruction, *Eur. Spine J.* **31**, 241–247 (2022), doi:10.1007/s00586-021-07024-7.
- ⁴⁵R. Vialle, N. Levassor, L. Rillardon, A. Templier, W. Skalli, and P. Guigui, Radiographic analysis of the sagittal alignment and balance of the spine in asymptomatic subjects, *J. Bone Joint Surg. Am.* **87**, 260–267 (2005), doi:10.2106/JBJS.D.02043.
- ⁴⁶P. J. Pickhardt, B. D. Pooler, T. Lauder, A. M. del Rio, R. J. Bruce, and N. Binkley, Opportunistic screening for osteoporosis using abdominal computed tomography scans obtained for other indications, *Ann. Intern. Med.* **158**, 588–595 (2013), doi:10.7326/0003-4819-158-8-201304160-00003.
- ⁴⁷M. G. Mody, A. Nourbakhsh, D. L. Stahl, M. Gibbs, M. Alfawareh, and K. J. Garges, The Prevalence of Wrong Level Surgery Among Spine Surgeons, *Spine (Phila. Pa. 1976)* **33**, 194–198 (2008), doi:10.1097/BRS.0b013e31816043d1.
- ⁴⁸N. Epstein, A perspective on wrong level, wrong side, and wrong site spine surgery, *Surg. Neurol. Int.* **12**, 286 (2021), doi:10.25259/SNI.402.2021.
- ⁴⁹E. C. Benzal, *Biomechanics of Spine Stabilization*, Thieme, New York, 3rd edition, 2015, ISBN 978-1-60406-924-2.
- ⁵⁰G. J. Hahn, W. Q. Meeker, and L. A. Escobar, *Statistical Intervals: A Guide for Practitioners and Researchers*, Wiley, Hoboken, NJ, 2nd edition, 2017.
- ⁵¹E. Beaulieu et al., A review of finite element analysis in spine surgery decision-making, *J. Clin. Med.* **15**, 2584 (2026), doi:10.3390/jcm15072584.